



Assertion-evidence slide template

Rodrigo Nemmen
Universidade de São Paulo

Assertion-evidence slide templates for scientific presentations

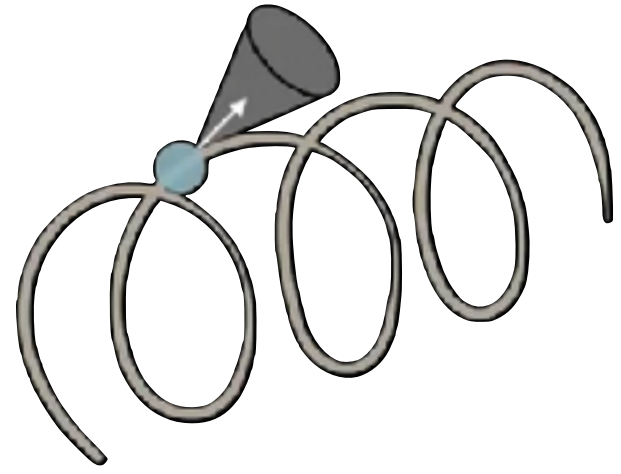
Updated Nov 2 2025 by R. Nemmen (Univ. de São Paulo)

- Based on the template by Michael Alley (Penn State)
- I (R. Nemmen) have been basing my presentations and university lectures on this template for almost two decades. It works.
- Copy this template and change it as you wish.

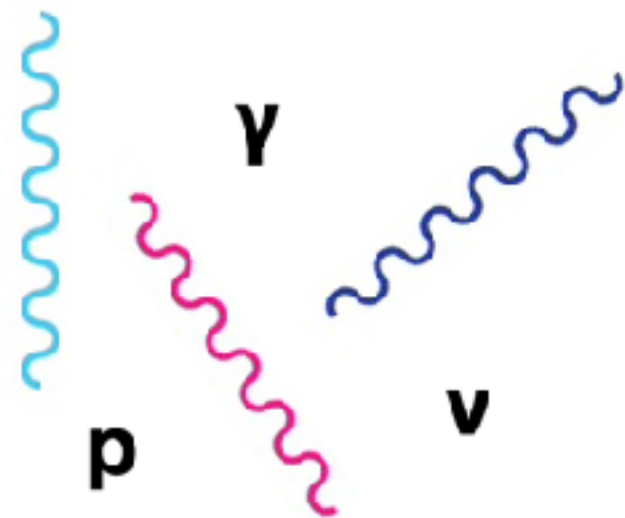
AGN jets

- **Formation**
- **Kinematics and dynamics**
- **Particle acceleration**
- **RL/RQ, FRI/FRII, blazar seq.**
- **Feedback**

Why are jets important?

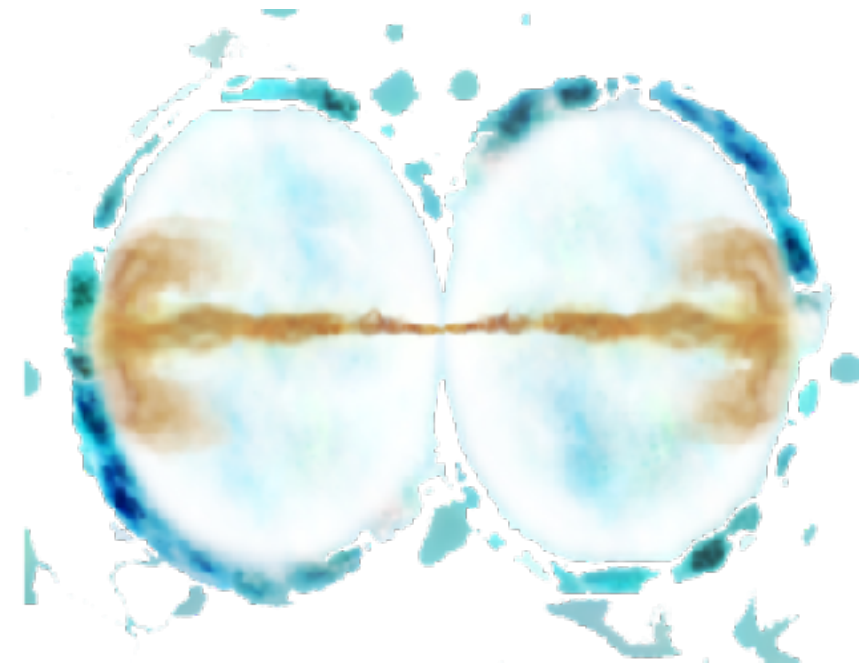


High-energy astrophysics and particle acceleration



Multimessenger astronomy:

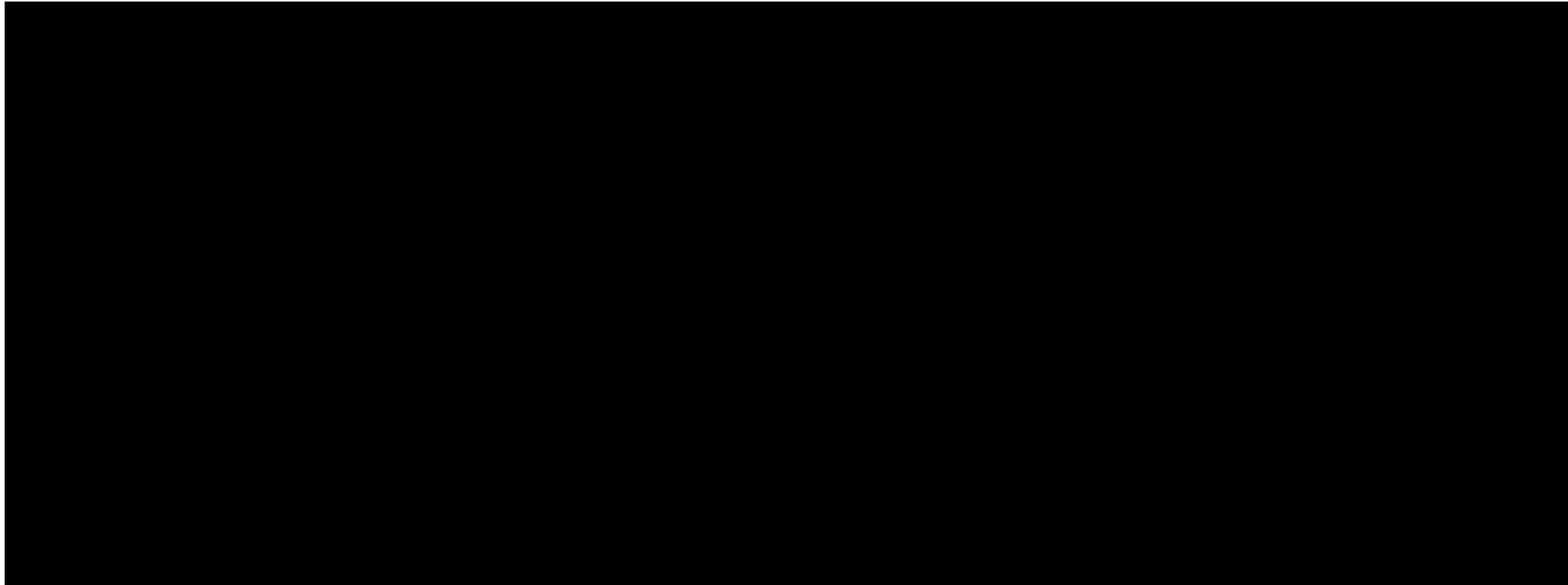
Sources of EM radiation, cosmic rays, neutrinos



Radio mode feedback:

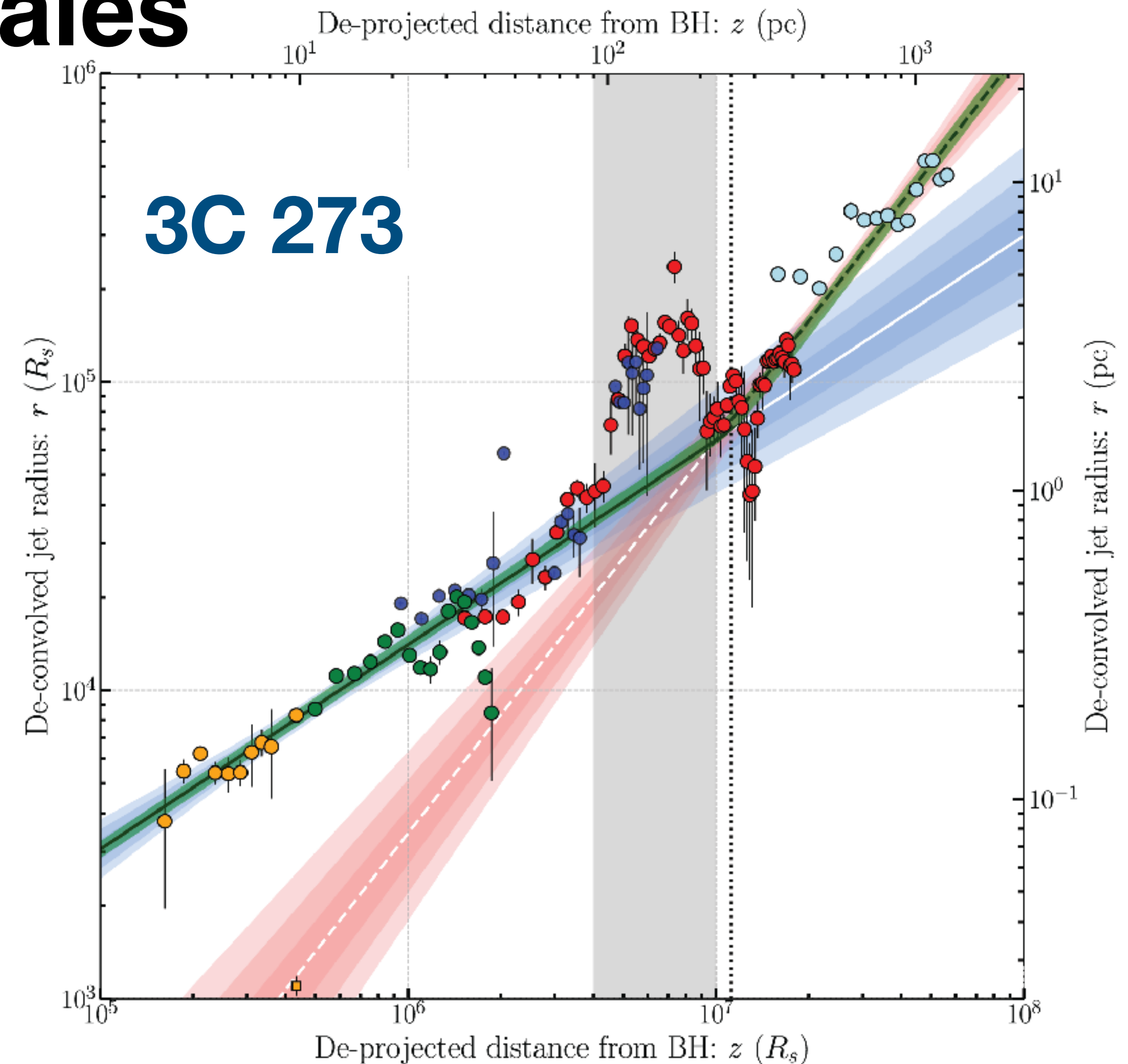
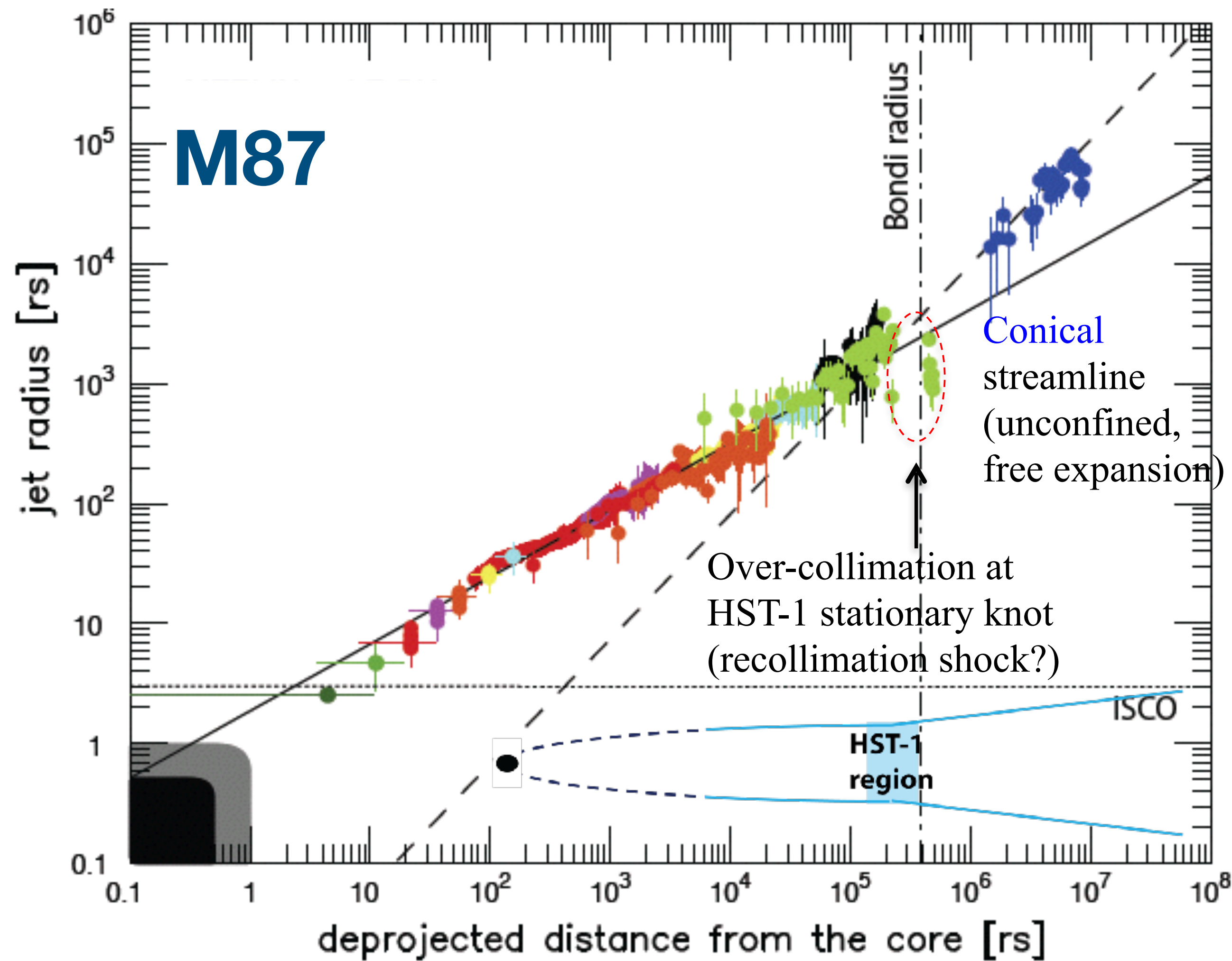
Impact large scale structure

Magnetic flux near BH is saturated in blazars and radio galaxies



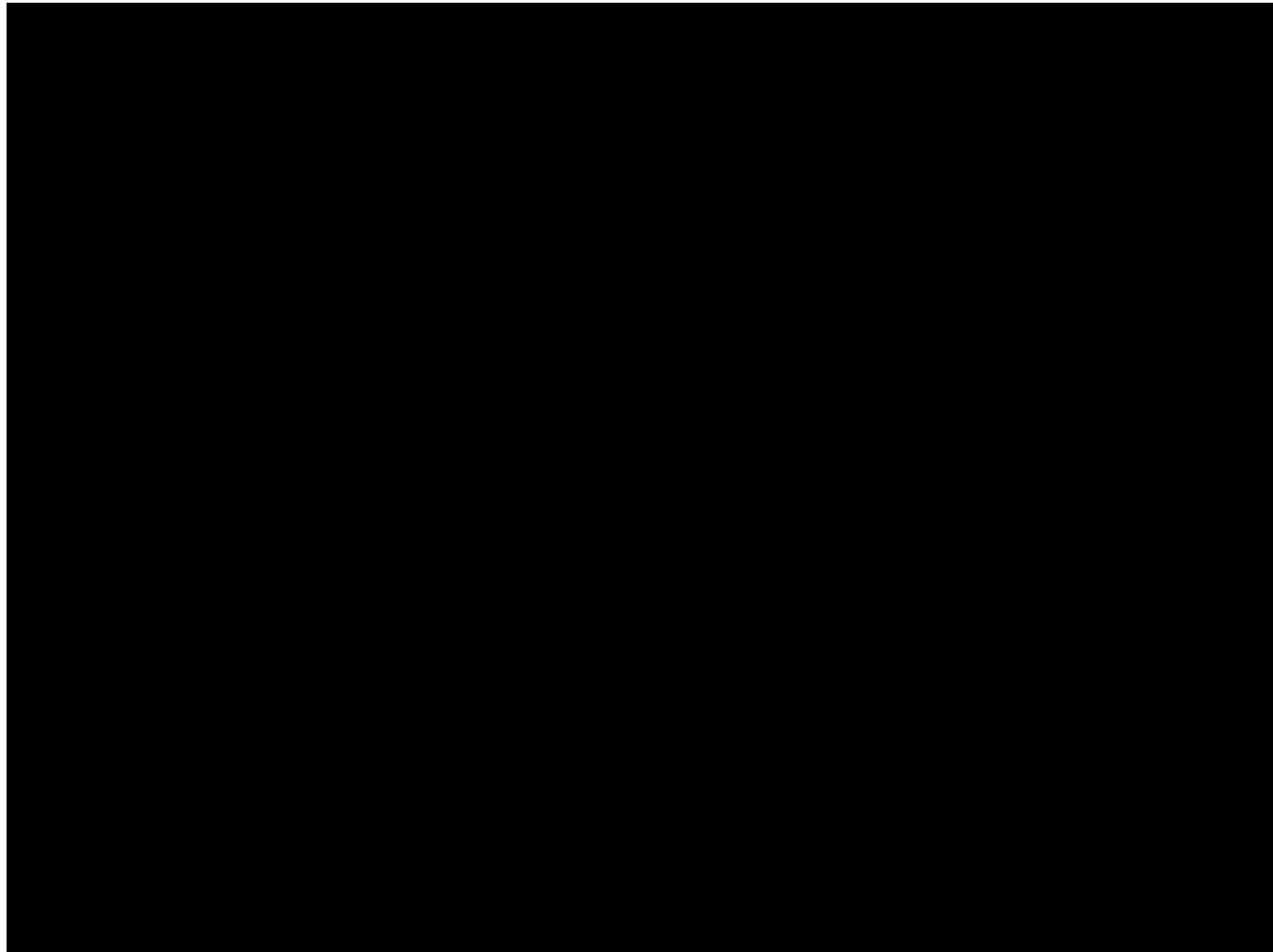
- Data from 47 numerical simulations spanning $q = 1$ to $q = 10$
- Deviation from linearity $< 3\%$ across entire range

Jets continue collimating through multiple mechanisms across vast scales



Asada & Nakamura 2012; Hada+2013; Okino+2022; M87 figure courtesy of Y. Mizuno

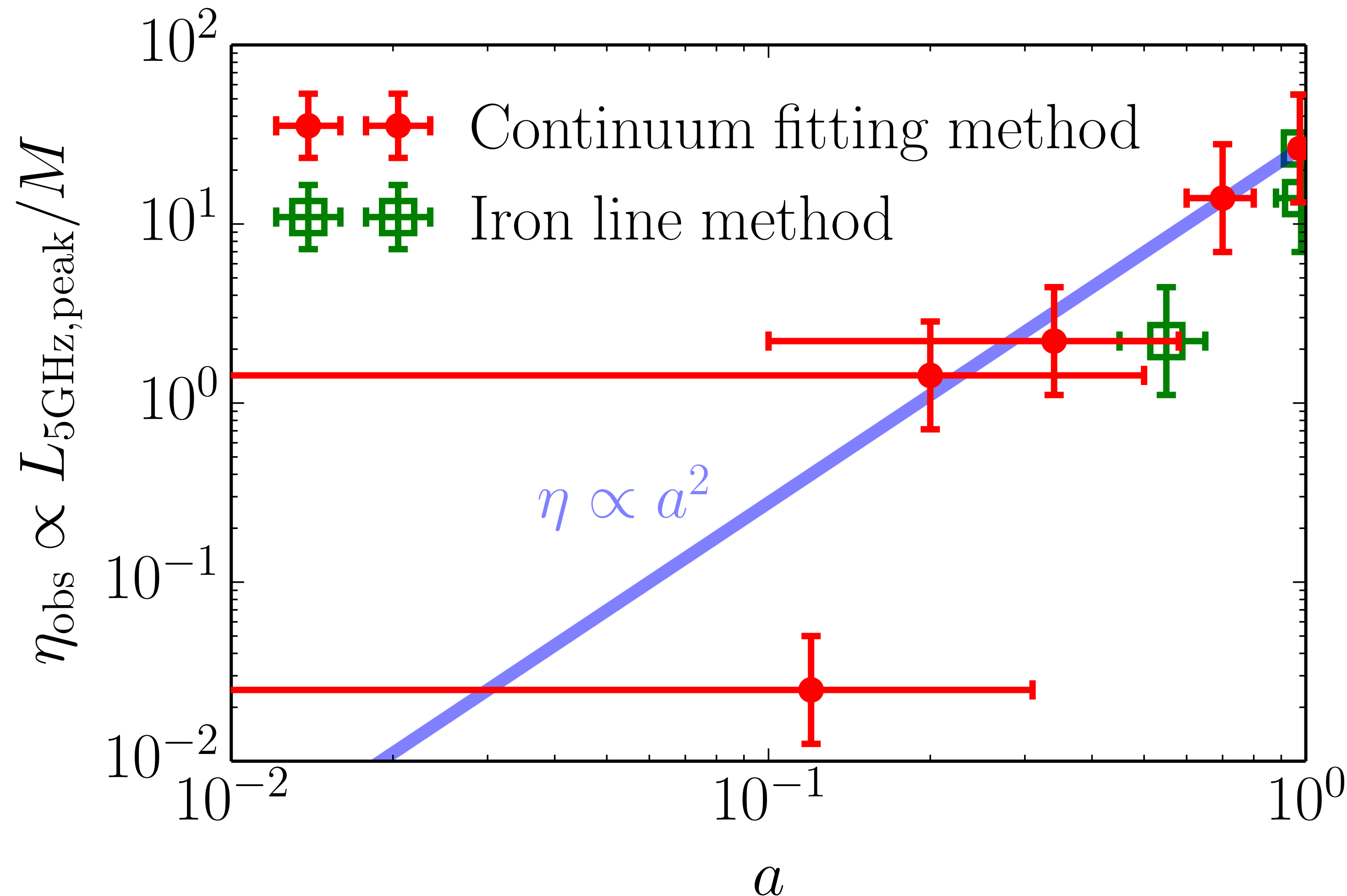
GR-PIC simulations show jet launching from first principles



- Spark gaps form near horizon (TeV flares), pair cascades from disk photons
- Refill magnetosphere w/ plasma
- BZ + Penrose process
- If seed photon field (accretion) too low to sustain cascades, no jet forms (relevant to RL/RQ)

Jet formation: Zou+2025; Yuan+2025; Niv+2023; **Parfrey+2019**; Other: Galishnikova+2023; Bransgrove+2022; Bacchini+2020; Crinquand+2020

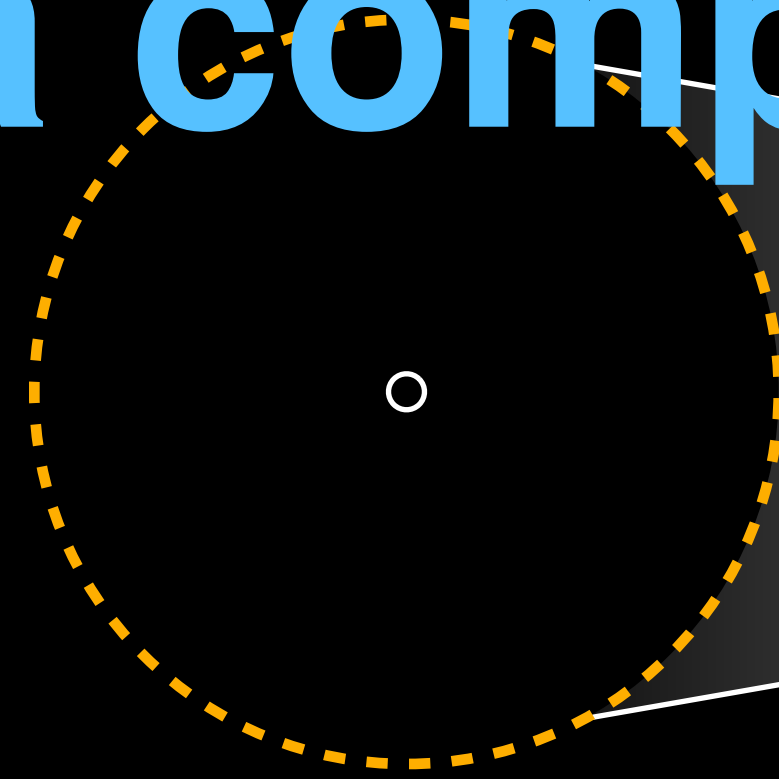
Stellar-mass BHs: Jet power seem to correlate with spin, but remains highly controversial



- Correlation holds for transient jets
- But not for steady jets (hard state)

Zona de influência gravitacional é minúscula comparada com a galáxia

Zona de influência gravitacional do BN



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